Half-Open Shell Certificates and Exact Colour Budgets

A scoped study of quantitative superexponential
bounds for van der Waerden numbers

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Abstract

We examine the September 23 source manuscript on quantitative superexponential van der Waerden bounds, keeping its torus counting, outer-colouring and geometric dichotomy as separate global inputs. A local sharpening uses the actual half-open squared-norm bands: a unit-separated line meets a unit band in at most two points, rather than the source's valid but weaker bound of four. This doubles the exponents in both perturbation failure estimates. Literal-key prescriptions give an exact finite probability ledger and a conditional-expectation certificate without introducing spurious subset-selection entropy. A mixed-base cyclic transfer retains all nonzero differences, and a standard periodic extension supplies the corresponding integer interval. Independently of the torus construction, we optimize the source's half-digit/sphere partition over integer parameters. The resulting logarithmic interval length has leading constant $1/(8\log 2)$, optimal for the stated colour-budget architecture, with an explicit lower-order loss and finite maximizer. Classical Behrend geometry, probability and digit arguments are credited; no independent certification of the source's two-colour superexponential theorem is asserted. Contribution assessment: C3/5, the sharp shell multiplicity and exact optimized colour budget are notable source-specific local refinements, not a new independently certified global growth theorem.

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1 Problem, status and the scope of the report

For integers $k, r \geq 2$, let $W_r(k)$ be the least positive L such that every map $[L] \to [r]$ has a monochromatic progression $a, a+d, \ldots, a+(k-1)d$ with $d > 0$. A colouring using fewer than r colours is allowed. A cyclic k-progression in $\mathbb{Z}/N\mathbb{Z}$ means the same sequence with $d \neq 0$ modulo N; its terms need not be distinct unless a further order condition is imposed. All unmarked logarithms are natural. In particular, three colours and the off-diagonal two-colour quantity traditionally called $w(3,k)$ are different problems.

The assigned source [1] asserts an absolute K_0 such that

$$W_r(k) > k^{10^{-5}k\lfloor \log_2 r \rfloor} \quad (k \geq K_0,\ r \geq 2). \tag{1}$$

The phrase *source asserts* is deliberate. This is stronger in the two-colour regime than a lower bound which only implies $W_2(k)/2^k \to \infty$. Reading the complete cached scientific text and checking the local deductions below do not constitute an independent certification of the global construction. Its asymptotic geometric and counting obligations are not discharged by any finite computation here.

1.1 Primary literature, checked October 8, 2026

Fox and Hunter's Theorems 1 and 2 [2], inspected in the primary full text, state $W_3(k) > 2^{k \log^* k/4}$ for sufficiently large k and, for each $0 < \varepsilon < 1$,

$$W_r(k) \geq r^{(1-\varepsilon)k \log k} \quad \text{if } k \geq k_0(\varepsilon), \quad r \geq (\log k)^{3/\varepsilon}.$$

Their least-nonzero-digit discussion also supplies historical context for digit products. Campos, Fox and Schildkraut's Theorem 1 [3] states $W_2(k) \geq (1 - o(1))k2^{k-1}$. Their Lemma 2 gives the periodic cyclic-to-interval extension used below; its short proof was inspected. These primary statements do not assert the source's all-r bound (1). Cao's September 25 abstract [4] claims $w(3, k) \geq k^{\Omega(\log k)}$, where the two forbidden lengths are 3 and k, not a diagonal three-colour bound. Only that abstract/metadata was inspected for this last reference. We do not claim an exhaustive novelty search or acceptance of every recent preprint proof.

1.2 What is original here, and what is not

The principal increment is the use of strict band endpoints in the source's actual perturbation keys, together with exact finite colour optimization in its independent all-colour construction. The probability ledger and mixed-base theorem make the interfaces auditable. They apply classical finite probability, conditional expectation and positional notation, not newly invented general methods. Behrend's sphere idea is classical [5]. The optimized leading constant below is optimal only under the specified, deliberately conservative colour-count formula. It is not an upper bound on $W_r(k)$, on all sphere constructions, or on the true number of distinct colours attained.

2 Complete source architecture and retained safeguards

The cached document comprises the main file, preamble, norm-band figure, eight scientific sections, bibliography and README. All thirteen texts were literally read in bounded displays. The map below separates the source's inputs to our conditional analysis from local results proved here.

Source module	Mechanism and retained obligation
01 Introduction	Distinguishes growth in forbidden length from growth in colour count; states a common k-threshold before all r.
02 Setup	Prime-power cyclic modulus; small-divisor-clearing multiplier; uniform and adaptive half-open meshes; endpoint and neighbour bounds.
03 Counting	Sign arrangements including zero signs; global word count; anchored local pattern count using literal boxes and constrained drifts.
04 Outer	Finite asymmetric local lemma on complete label variables; light-word row balancing and eligible long-block balancing.
05 Dichotomy	Small-box real affinity, primitive return, short-return cut exclusion and long-block coverage; rich outer colours or a global real affine lifted progression.
06 Perturbation	Independent flips indexed by literal box and norm band; bounded key multiplicity; quadratic floor-word count and two separate union estimates. Our sharp factor two enters here.
07 Transfers	Least-nonzero-digit transfer, uniform all-r quantifier; separate half-digit/sphere partition. Our integer optimization enters only the latter partition.
08 Upper	Focusing induction for qualitative finiteness, with recursively enlarged colour alphabets. Not a lower-bound construction or a verification of a companion quantitative upper bound.

2.1 Parameter and normalization ledger

The source uses, for sufficiently large integer k,

$$D = \lceil k^{1/10} \rceil, \quad M = \lceil k^{1/2} \rceil, \quad h_0 = D^2, \quad H = k^{-2}, \quad \eta = (1000D)^{-1},$$
$$p = k^{-1/20}, \quad \gamma = 1/100, \quad \delta = 1/10, \qquad \lambda = \prod_{\ell \le h_0,\ \ell \text{ prime}} \ell^{\lfloor \log(2M)/\log \ell \rfloor}. \tag{2}$$

An elementary prime-existence argument supplies $k < P \le 2k^2$; q is the least power of P at least $k^{ck/D}$, and $N = q^D$. For fixed $c > 0$,

$$k^{ck/D} \le q \le 2k^2 k^{ck/D}, \qquad \log N \le ck \log k + D \log(2k^2). \tag{3}$$

All primes in λ are below P, so λ is a unit modulo N. Every nonzero cyclic difference has order at least $P > k$; this particular construction has distinct progression terms.

For $n \in \mathbb{Z}/N\mathbb{Z}$, put $x_i(n) = n/q^i \pmod 1$ and $y_i(n) = \lambda x_i(n)$, $1 \le i \le D$. Use representatives $\widehat{x} \in [0,1)^D$ and $\widetilde{y} \in [-1/2, 1/2)^D$. The x-mesh has width H/λ. The adaptive y-mesh uses a cut-distance ρ, a positive floor $\alpha = q^{-(D+1)}$, and intervals comparable to $H(\rho + \alpha)$. Its intervals are half-open. A full label $L(n)$ contains the literal locations of all x- and y-intervals. A truncated rotating coordinate records its full interval only when $\rho \ge \eta$. The outer assignment c_* is fixed on full labels, and $c_0(n) = c_*(L(n))$.

2.2 Counting and outer-colouring interfaces

The global word count has

$$\log T_{\text{glob}} = O(D(\log k + \log \lambda)) = O(k^{3/10} \log k) = o(M).$$

The local count through a specified full label, for $h_0 < h \le 2M$, is bounded by $(CDh)^{6D}$. The source includes zero hyperplane signs and anchors a regular position to its actual box: it does not choose an arbitrary regular mask or identify boxes solely by their shape. These details keep an uncontrolled mesh-size factor out of the local count.

The outer local lemma uses independent bits on full labels before any norm-band perturbation. Light labels occur at most k/M times in a progression. Splitting their occurrences into rows with no repeated label yields long rows on which a Chernoff estimate applies. For eligible long-return blocks, at least half the positions are regular and their labels are pairwise distinct. The source budgets event weights by the number of exposed labels, with a small uniform per-label neighbour sum. Independence from the sigma-algebra of non-neighbour events is essential; pairwise independence is insufficient.

The geometric dichotomy is: every nonzero-step k-progression either has at least γk positions of each outer colour, or its entire lifted y-sequence is real affine. The source uses return arithmetic and cut control, not an assumption that a torus progression lifts affinely. A small return divides λ; a segment meeting a cut is treated even when the cut is an endpoint. Long-return blocks retain stationary-coordinate drift bounds obtained from a heavy visit, even if that visit is not regular. These valid details are not removed by our local modification.

2.3 A conditional interface, not an audit shortcut

In Sections 3–4, assume this fixed colouring, dichotomy, geometric affinity and counting with their stated uniformity. We prove how they enter the sharpened completion. They have not thereby

been derived from a finite matrix or certified in every parameter. The outer local lemma, complete anchored arrangements, all cut cases, effective thresholds and global dichotomy remain source obligations. The companion quantitative upper-bound theorem and complete proofs of external references are not audited here.

3 The sharp half-open shell bound

Lemma 3.1 (Two points, not four)**.** *Let A be a finite subset of a real affine line in a Euclidean space, with distinct points at distance at least $\sigma > 0$. For any real u and $0 < w \le \sigma^2$,*

$$\#\{z \in A : u \le \|z\|^2 < u + w\} \le 2.$$

The constant two is sharp already in dimension one.

Proof. Write $z = z_* + te$, where $\|e\| = 1$ and $z_* \perp e$. Then $\|z\|^2 = \|z_*\|^2 + t^2$ and distance is $|t - t'|$. If $0 \le t < t'$ are on the same ray, $t' - t \ge \sigma$, so

$$(t')^2 - t^2 = (t' - t)(t' + t) \ge \sigma^2.$$

Two values in the same half-open interval of length $w \le \sigma^2$ have difference strictly less than σ^2, a contradiction. The negative ray is identical after $t \mapsto -t$. Assign $t = 0$, if present, to just one ray. There is at most one point on each ray. For sharpness, $t = \pm 3/5$ have distance $6/5 \ge 1$, with both squares in $[0, 1)$. □

Half-open endpoints matter. The three unit-separated points $-1, 0, 1$ all have squares in the closed band $[0, 1]$. Widening the half-open band to $[0, 1 + \varepsilon)$, $\varepsilon > 0$, also admits all three. A rounding rule changing the upper endpoint can invalidate the multiplicity. The source's bound four follows by enclosing each ray piece in a closed interval of length one; it is a valid relaxation, not a contradiction in the source proof. We retain the strict band and avoid that loss.

3.1 Application to the actual source keys

The source flip key is

$$\kappa(n) = \big(V_1(y_1(n)), \dots, V_D(y_D(n)), \lfloor f(n) \rfloor\big), \qquad f(n) = \|q\widetilde{y}(n)\|^2. \tag{4}$$

The full x-label is absent. All points with this literal key share one flip, even if their outer colours differ.

Proposition 3.2 (Source-key multiplicity)**.** *Assume the source's small-box affinity conclusion: lifted points of a fixed y-box along a nonzero-step k-progression lie on a real affine line. Under* (2)–(3)*, each key* (4) *occurs at most twice on that progression.*

Proof. Any two distinct cyclic terms have scaled lifted distance at least one. Write the nonzero residue of their λ-multiplied difference modulo q^D as $q^a u$, where $0 \le a < D$ and $q \nmid u$. Coordinate $a + 1$ has displacement $u/q \pmod 1$, whose circular distance from zero is at least $1/q$. Ordinary representative distance is no smaller; multiplication by q proves separation. This uses divisibility by the integer q, not an incorrect assertion that q itself is prime.

Positions with a fixed key lie in a fixed y-box, hence on the stipulated line, and in the same half-open unit squared-norm band. Apply Lemma 3.1 with $\sigma = w = 1$. □

The affinity mechanism requires its exact size premise. For indices $j_1 < j_2 < j_3$ in one box of coordinate width at most $2H$, the vector

$$(j_3 - j_2)(Y_{j_2} - Y_{j_1}) - (j_2 - j_1)(Y_{j_3} - Y_{j_2})$$

is integral because the underlying points form a torus progression. Each coordinate has absolute value at most $2kH < 1$, so it vanishes. This proves affinity on these indices, including nonconsecutive visits. Distinctness and separation prevent a constant line when two or more visits occur.

3.2 Exact label stability, not merely a small residual

If an implementation encloses $f(n)$ by a rational interval $[\underline{f}, \overline{f}]$, it can certify the floor m when $\underline{f} \geq m$ and $\overline{f} < m + 1$, or by an equivalent exact endpoint proof. A small absolute error alone does not decide the key near an integer. The analogous requirement holds at every adaptive box boundary with its half-open convention. Exact boundary values may need symbolic equality; no uniform positive clearance is promised. Refining enclosures or retaining multiple possible labels is legitimate, but silently substituting a nearby key is not. The factor two is a theorem about exact keys, not a numerical observation about approximate buckets.

4 Exact prescriptions and the sharpened completion

Let $\mathcal{K}$ be the finite set of literal keys. Fix the outer colouring first, then take independent Bernoulli variables $(E_\kappa)_{\kappa \in \mathcal{K}}$ with $\mathbb{P}(E_\kappa = 1) = p$, $0 < p \leq 1/2$. The perturbed colour is

$$C(n) = c_0(n) + E_{\kappa(n)} \pmod 2.$$

No independence between distinct progressions is needed.

Proposition 4.1 (Exact event mass)**.** *For a specified progression, if a key occurs with both outer colours, the probability of a monochromatic final progression is zero. Otherwise let U be the set of its keys with outer colour one and V the set with outer colour zero. Then $U \cap V = \varnothing$, and the probability of monochromaticity equals*

$$p^{|U|}(1-p)^{|V|} + p^{|V|}(1-p)^{|U|}. \tag{5}$$

If every key has multiplicity at most two and the progression has k positions, its probability of being identically a specified colour is at most $(1-p)^{\lceil k/2 \rceil}$. If at least γk positions have the opposite outer colour, that probability is also at most $p^{\lceil \gamma k/2 \rceil}$.

Proof. For target colour zero the required assignment is $E = 1$ on U and $E = 0$ on V; for target one it is the complement. A key with conflicting outer colours would require both bit values. Otherwise independence gives the two displayed monomials. The target events are disjoint for a nonempty progression. At least $\lceil k/2 \rceil$ keys occur. Each prescribed bit has probability at most $1 - p$. Likewise the opposite-colour positions force at least $\lceil \gamma k/2 \rceil$ different keys to flip. Keeping only these necessary flips gives the second bound. □

The forced flip set is determined by the fixed progression and its outer colours. There is no choice of a subset of positions to union over, hence no binomial entropy factor. Conversely, replacing literal keys by anonymous equality patterns is unsafe: identical patterns on different keys need not define the same random event.

4.1 A finite failure budget

Suppose a complete family of cyclic progressions is covered by rich progressions as above and by affine signatures, where a signature records the full ordered list $(c_0(n_j), \kappa(n_j))_{j=0}^{k-1}$. Let T bound the number of distinct literal affine signatures. The two families may overlap.

Theorem 4.2 (Sharpened finite completion)**.** *Under the key-multiplicity and covering premises, if*

$$2N(N-1)p^{\lceil \gamma k/2 \rceil} + 2T(1-p)^{\lceil k/2 \rceil} < 1, \tag{6}$$

some assignment of the key bits gives a colouring of $\mathbb{Z}/N\mathbb{Z}$ with no monochromatic nonzero-step k-progression.

Proof. There are $N(N-1)$ ordered choices of initial term and nonzero difference. For each rich progression and each target colour use Proposition 4.1. For affine progressions use one event for each literal signature and target; equal signatures give exactly equal events on the same probability space. The union bound gives (6). Double-counting a progression in both families only increases the bound. If that bound is below one, the complement of all bad events has positive probability. □

The source already counts affine signatures through the quadratic form of $f(n_j)$. On a globally real affine lifted progression,

$$f(n_j) = \theta_0 + \theta_1 j + \theta_2 j^2, \qquad 0 \le f(n_j) \le Dq^2/4.$$

Comparisons with all relevant integer levels form $O(k(Dq^2+2))$ hyperplanes in three coefficient parameters. The arrangement count, including equality signs, gives

$$T \le CT_{\mathrm{glob}}\,[k(Dq^2+2)]^3, \qquad \log T = O(k^{3/10}\log k) + 6\log q + O(\log k). \tag{7}$$

Full-label words determine the outer colours; the recorded y-boxes and floor values determine the literal keys. This is why the three-parameter count can be combined with the global word count. We retain the source's arrangement input, rather than asserting that every abstract quadratic sequence is geometrically realized.

Corollary 4.3 (Conditional asymptotic budget)**.** *Assume that the source's entire upstream interface and* (7) *remain valid for a fixed $c > 0$ in* (3)*. With the other parameters in* (2)*, the final budget tends to zero whenever*

$$0 < c < \gamma/80.$$

This is a sufficient range for the completion module, not an independent global theorem with a new constant.

Proof. Using ceilings only strengthens the estimates, and $\log p = -(\log k)/20$. The logarithm of the rich contribution is at most

$$\log 2 + (2c - \gamma/40)k\log k + 2D\log(2k^2). \tag{8}$$

The final term is $o(k\log k)$, so the stated strict inequality makes this tend to minus infinity. For the affine contribution,

$$\log(2T) - p\lceil k/2 \rceil \le \log(2T) - \tfrac{1}{2}k^{19/20} \longrightarrow -\infty,$$

since $\log q = O_c(k^{9/10}\log k)$. Add the two bounds. □

With multiplicity four, the corresponding rich coefficient was $2c - \gamma/80$, requiring $c < \gamma/160$. For the source's actual $c = 10^{-5}$, our coefficient is $-23/100000$, compared with its $-21/200000$. The affine suppression changes from $e^{-pk/4}$ to $e^{-pk/2}$. For illustration $c = 10^{-4} < \gamma/80 = 1/8000$ passes the new *final* budget. We do not insert it into a newly asserted global bound: every upstream uniform estimate and parameter choice would have to be justified for that changed input, and an effective K_0 has not been calculated.

4.2 A tighter literal-event certificate

The ordered signature count is sufficient, but a finite implementation can discard more duplication without probabilistic assumptions. For every consistent progression form the unordered pair $\{(U, V), (V, U)\}$ of disjoint literal key sets. Choose one representative for each distinct pair, obtaining a set $\mathcal{Q}$. Then

$$\Psi(p) = \sum_{\{U,V\}\in\mathcal{Q}} \left[p^{|U|}(1-p)^{|V|} + p^{|V|}(1-p)^{|U|}\right] < 1 \tag{9}$$

is a complete avoidance certificate, provided $\mathcal{Q}$ accounts for *all* nonzero-step progressions. This is an exact sum of individual bad-event probabilities, not the exact union probability; different prescriptions can overlap. The quotient is by literal sets, not by their cardinalities. It may reduce a concrete certificate but gives no proved asymptotic improvement on the source's signature count.

For unequal independent probabilities $0 < p_\kappa < 1$, use

$$\prod_{\kappa\in U} p_\kappa \prod_{\kappa\in V} (1 - p_\kappa)$$

and its complementary prescription. This extension is exact; uniform bounds such as (6) require their own extremal-probability estimates.

Proposition 4.4 (Deterministic extraction from the finite ledger)**.** *If a complete finite prescription list has sum of event probabilities less than one, its independent key variables can be fixed one at a time so that no listed event occurs.*

Proof. Let Z count listed bad target prescriptions. Conditional on the already fixed keys, its expected value is a sum of surviving prescription monomials. For the next key this expectation is the weighted average of the two conditional values, so at least one choice does not increase it. At the end Z is a nonnegative integer strictly below one, hence zero. Independence supplies the product weights for the remaining keys; no event independence is used. □

For example, at $p = 1/3$, the prescriptions $(E_0, E_1) = (1, 1)$ and $(E_1, E_2) = (0, 1)$ have total mass $1/9 + 2/9 = 1/3$. Fixing $E_0 = 0, E_1 = 1, E_2 = 0$ gives the conditional ledger $1/3, 2/9, 0, 0$. This miniature witness checks the extraction algebra; it is not a large-k cyclic colouring. Building the complete prescription list can cost at least the enumeration of $N(N-1)$ progressions and many key incidences. The result promises neither polynomial time in $\log N$ nor an efficient construction of the outer colouring.

5 Cyclic products and complete integer intervals

Theorem 5.1 (Mixed-base cyclic product)**.** *Fix $k \geq 2$. For $0 \leq i < m$, suppose $N_i \geq 2$ and $C_i : \mathbb{Z}/N_i\mathbb{Z} \to [r_i]$ avoids every nonzero-step cyclic k-progression. Put*

$$B_0 = 1, \quad B_i = \prod_{j<i} N_j, \quad B = B_m, \quad R = \prod_{i<m} r_i.$$

The digit-tuple colouring is cyclically k-progression-free on $\mathbb{Z}/B\mathbb{Z}$, even when the bases are not coprime. It yields

$$W_R(k) > (k-1)B. \tag{10}$$

Proof. For a residue $n \pmod B$, digit i is $\lfloor n/B_i \rfloor \pmod{N_i}$, well defined independently of the integer representative, because $B_i N_i \mid B$. Assign the tuple of its C_i-colours. For $0 < d < B$, choose the largest $i < m$ such that $B_i \mid d$. Since these prefixes form a divisibility chain and $B \nmid d$, $N_i \nmid e = d/B_i$. For every integer a, j,

$$\left\lfloor \frac{a + jd}{B_i} \right\rfloor = \left\lfloor \frac{a}{B_i} \right\rfloor + je.$$

Reducing modulo N_i produces a cyclic progression of nonzero step in digit i. The tuple cannot be monochromatic. Reduction of $a + jd$ modulo B changes the quotient by a multiple of N_i, so wrapping around the product cycle causes no problem.

Now repeat this colouring periodically on $\{0, \ldots, (k-1)B - 1\}$. Any integer k-progression in this interval has $0 < d \leq B - 1$, since $(k-1)d \leq (k-1)B - 1$. Its residues contradict cyclic avoidance. Translation by one gives the required colouring of $[(k-1)B]$. □

The periodic factor $k - 1$ is the classical observation in Campos–Fox–Schildkraut [3, Lemma 2]; its use after the mixed-base product is not credited as a newly discovered principle. The assigned source uses equal bases and restricts to length N^m. Its least-nonzero-digit proof is correct. The theorem above records the slightly stronger cyclic closure and the available periodic length explicitly.

Repeated use of one two-colour cyclic seed of size N, with $m = \lfloor \log_2 r \rfloor$, gives

$$W_r(k) > (k-1)N^{\lfloor \log_2 r \rfloor} \quad (r \geq 2).$$

If the seed exists for every $k \geq K_0$ with one absolute threshold, the same threshold works for every r, however large. No limiting argument uniform in an unbounded digit count is needed: each finite number of digits uses the same seed and an exact identity.

Example 5.2 (Small, complete certificates)**.** The colourings $C_3 = (0, 0, 1)$ and $C_4 = (0, 0, 1, 1)$ avoid all nonzero-step cyclic 3-progressions, as checked directly. They give a four-colouring of the product cycle of size 12, and then an avoiding four-colouring of an interval of length 24. The order-two difference in $\mathbb{Z}/4\mathbb{Z}$ repeats a term, but still has both colours. Ignoring repeated-term cyclic progressions would leave precisely this kind of transfer obligation unchecked.

An interval seed alone is not enough. Take $(0, 0, 1, 0)$ on $\{0, 1, 2, 3\}$: both integer three-progressions are nonmonochromatic, whereas the cyclic progression $3, 0, 1$ is all zero. Its periodic continuation therefore fails on $3, 4, 5$. Thus one must test the complete cyclic predicate, not substitute an interval-only search.

6 Exact optimization of the all-colour partition

This section does not assume any torus construction or outer local lemma. Its fully explicit colouring works for every $k \geq 3$. We reproduce the short geometric proof so the quantitative change is independent of a citation to the source's coarse estimate.

Proposition 6.1 (Half-digit and shell colouring)**.** *For integers $b \geq 2$, $s \geq 1$, an interval of length b^s has a colouring with at most*

$$Q(b,s) = 2^s\big(1 + s(b-1)^2\big) \tag{11}$$

colours and no nonconstant three-term progression.

Proof. Write $n = \sum_{i=0}^{s-1} x_i b^i$, $0 \leq x_i < b$, for $0 \leq n < b^s$. Partition the digit set into consecutive halves

$$H_0 = \{0, \dots, \lceil b/2 \rceil - 1\}, \qquad H_1 = \{\lceil b/2 \rceil, \dots, b-1\}.$$

Let $h(x_i)$ record the half. Give n the colour

$$\left((h(x_i))_{i=0}^{s-1},\ \sum_{i=0}^{s-1} x_i^2 \right). \tag{12}$$

The norm is an integer between zero and $s(b-1)^2$, proving the colour bound.

If n_x, n_y, n_z form a monochromatic three-progression, their digit vectors are in the same half in every coordinate. Consequently $a_i = x_i - 2y_i + z_i$ satisfies

$$|a_i| \leq 2(\lceil b/2 \rceil - 1) < b, \qquad \sum_{i=0}^{s-1} a_i b^i = 0.$$

Reducing modulo b gives $a_0 = 0$; division by b and induction give $a_i = 0$ for every i. Hence $y = (x+z)/2$ in the actual Euclidean digit space. All three squared norms are equal, say Q_0. The parallelogram identity yields

$$Q_0 = \|(x+z)/2\|^2 = \tfrac{1}{2}(\|x\|^2 + \|z\|^2) - \tfrac{1}{4}\|x - z\|^2,$$

so $x = z = y$, a constant progression. Translation by one completes the interval colouring. □

This is the source's construction, retaining both the no-carry half information and the exact integer shell, based on classical Behrend geometry. We improve its parameter choice, not the geometry.

Theorem 6.2 (Exact finite budget and an explicit choice)**.** *For an integer $r \geq 4$, define the finite admissible set*

$$\mathcal{S}(r) = \{s \geq 1 : 2^s(1+s) \leq r\}, \qquad b_s = 1 + \left\lfloor \sqrt{\frac{\lfloor r/2^s \rfloor - 1}{s}} \right\rfloor,$$

and

$$L_*(r) = \max_{s \in \mathcal{S}(r)} b_s^s. \tag{13}$$

Then $W_r(k) > L_*(r)$ for every $k \geq 3$. The maximum is the largest interval length certified by (11)*, over all integer choices $b \geq 2, s \geq 1$.*

In particular set

$$s_0 = \lfloor (\log_2 r)/2 \rfloor, \quad B_0 = \lfloor r/2^{s_0} \rfloor, \quad b_0 = 1 + \left\lfloor \sqrt{(B_0 - 1)/s_0} \right\rfloor. \tag{14}$$

These parameters have $s_0 \geq 1, b_0 \geq 2, Q(b_0, s_0) \leq r$. *For* $r \geq 16$, *with* $L = \log r$ *and* $a = \log 2$,

$$\log L_*(r) \;\geq\; \log b_0^{s_0} \;\geq\; \frac{L^2}{8a} - \frac{L}{4a} \log\left(\frac{L}{a}\right) - \frac{a}{2}. \tag{15}$$

Proof. For a fixed s, condition $Q(b, s) \leq r$ is equivalent to

$$(b-1)^2 \leq (\lfloor r/2^s \rfloor - 1)/s.$$

Its largest integer solution is b_s, and $b_s \geq 2$ exactly when $2^s(1+s) \leq r$. The admissible set is nonempty for $r \geq 4$, and $s < \log_2 r$, so taking the finite maximum proves optimality within this budget. Proposition 6.1 avoids three-progressions and therefore every k-progression with $k \geq 3$; embed the colour labels into $[r]$.

For (14), $r \geq 2^{2s_0}$, so $B_0 \geq 2^{s_0} \geq s_0 + 1$. Thus $b_0 \geq 2$, and its defining floor guarantees the budget. For $r \geq 16$ we also have $r/2^{s_0} \geq \sqrt{r} \geq 4$, and hence

$$B_0 - 1 \geq r/2^{s_0} - 2 \geq r/(2^{s_0+1}).$$

Since $1 + \lfloor x \rfloor \geq x$ for $x \geq 0$,

$$\log b_0 \geq \tfrac{1}{2}(L - s_0 a - \log(2s_0)).$$

Write $s_0 = L/(2a) - t$, where $0 \leq t < 1$. Then

$$\frac{s_0 L}{2} - \frac{a s_0^2}{2} = \frac{L^2}{8a} - \frac{a t^2}{2} \geq \frac{L^2}{8a} - \frac{a}{2}.$$

Finally $s_0 \leq L/(2a)$ and $\log(2s_0) \leq \log(L/a)$, both positive in this range. Multiplication gives (15). □

The source's displayed all-colour bound uses $1/(64 \log 2)$ with its more conservative choices and estimates. Our optimized budget has leading coefficient $1/(8 \log 2)$, with the explicit negative $L \log L$ term retained. Dropping that term would not give a proved finite lower bound. Neither comparison claims the best historical constant for all multicolour constructions.

Proposition 6.3 (Leading optimality within this architecture)**.** *For every admissible pair* $b \geq 2, s \geq 1$ *with* $Q(b, s) \leq r$,

$$\log(b^s) \leq \frac{(\log r + 2 \log 2)^2}{8 \log 2}. \tag{16}$$

Consequently

$$\lim_{r \to \infty} \frac{\log L_*(r)}{(\log r)^2} = \frac{1}{8 \log 2}.$$

Proof. The budget implies $2^s s(b-1)^2 \leq r$. Use $b \leq 2(b-1)$, take logarithms and multiply by s:

$$s \log b \leq \tfrac{s}{2}(L - sa - \log s) + sa \leq \tfrac{s}{2}(L + 2a) - \tfrac{a}{2} s^2 \leq \frac{(L+2a)^2}{8a}.$$

The second inequality uses $\log s \geq 0$, and the last is completion of the square. Combine with (15) and divide by L^2. □

6.1 Finite examples and the cost of exact optimization

r	b_0, s_0	$b_0^{s_0}$	optimal b, s	$L_*(r)$	$Q(b,s)$
16	$2,2$	4	$2,2$	4	12
64	$2,3$	8	$3,2$	9	36
256	$2,4$	16	$4,3$	64	224
1024	$3,5$	243	$7,3$	343	872
65536	$6,8$	1679616	$14,6$	7529536	64960

These are exact integer outputs, not floating-point searches. For example $r = 256, b = 4, s = 3$ uses at most $8(1 + 3 \cdot 9) = 224$ colours on length 64. The source's original choice at that r is $b = 4, s = 2$, giving length 16. Actual attained colour counts can be less than $Q(b, s)$; the table optimizes the guaranteed budget, not those smaller counts.

There are fewer than $\log_2 r$ candidate dimensions. Each uses integer division, integer square root, exponentiation and comparison. This describes a small finite parameter search; it is not a unit-cost bit-complexity claim for producing the entire colouring. Evaluating the colour of one integer uses its s digits and a sum of squares. Printing all b^s colours necessarily has output cost at least that many entries. Exploiting the actually attainable shell values or replacing the two halves could improve the budget, but requires a new no-carry proof and a new count; it is outside Proposition 6.3.

7 Original Ram comparisons and certificate discipline

The confidential comparison TR-item-233-astra [6] was read in full, and relevant original derivations in TR250 [7] were freshly read rather than inferred from that comparison. These inputs suggest an exact feasible-set discipline, not a new source of interval colourings.

7.1 Property B and the quantifier that must not change

For a finite integer set S, let $\mathcal{H}_k(S)$ contain every k-element arithmetic progression in S. For spins $\sigma_v \in \{-1, 1\}$ and an edge $e = \{v_0, \dots, v_{k-1}\}$, the classical indicator identity is

$$M_e(\sigma) = 2^{1-k} \prod_{i=1}^{k-1} (1 + \sigma_{v_0}\sigma_{v_i}). \tag{17}$$

It equals one exactly on monochromatic edges and zero otherwise: each factor is two or zero. Thus $\sum_{e \in \mathcal{H}_k(S)} M_e(\sigma) = 0$ is exactly an avoiding colouring on the specified complete hypergraph. For an interval of length v, the number of edges is

$$\sum_{d=1}^{q} \big(v - (k-1)d\big) = qv - \tfrac{1}{2}(k-1)q(q+1), \qquad q = \lfloor (v-1)/(k-1) \rfloor.$$

This count is a useful completeness check for a finite enumerator.

Sparse forcing asks whether every colouring of some specially chosen S has an edge; interval avoidance asks for a colouring of every point of $[v]$ with no edge. Bounds for the least cardinality of a forcing S, or density/degree estimates for sparse hypergraphs, do not supply a colouring of a long interval. The corresponding outer existential/universal choices cannot be interchanged. Nor does a colouring avoiding a sampled edge list certify the complete interval predicate.

7.2 What the original finite-coordinate proof contributes

TR250's original Sections on physical spaces, compatible coordinates and constrained acceptance explicitly retain the actual feasible space under a change of coordinates. If $\sigma = Rm$ with R invertible, the spin constraints become

$$m \in R^{-1}\{-1,1\}^{|S|}, \qquad \sum_{e \in \mathcal{H}_k(S)} M_e(Rm) = 0.$$

Both the objective and the domain must be transformed. A ball, a linear null space or an arbitrary collection of moment constraints is not this discrete set. This follows simply by substitution and bijectivity; no new Ramsey theorem is credited to it. Similarly, the original preservation of compatible kernels and quotients under invertible maps does not assert that a finite map has a uniform lower bound on an infinite family.

The useful source-specific application is to the literal prescription ledger: a certificate records its actual key universe, the complete progression constraints, exact endpoint decisions and precise probability weights. A compressed representation must reconstruct those data, not only small residuals of a surrogate system. Conditional expectation in Proposition 4.4 works because it retains the exact discrete domain at every step. Neither the original Ram proof nor this transport observation constructs the source's torus meshes or proves its outer local lemma.

8 Limits, advantages and next questions

Advantages. The shell improvement is insensitive to dimension and uses the exact band already present in the source. It doubles both relevant probability exponents without a changed mesh or a new probabilistic lemma. The finite ledger can exploit inconsistent and duplicate prescriptions, preserving literal dependencies. Mixed bases allow seeds of different sizes and keep all nonzero differences, while the all-colour partition offers a separate, elementary lower bound with an exact finite parameter optimizer.

Costs and limitations. Key boundaries require exact arithmetic or certified endpoint logic. The factor two is sharp for the geometric assumptions, so further improvement needs structure beyond a separated line in a unit shell. The ledger can be huge; the outer colouring remains an imported existence argument. No effective absolute threshold K_0, no large source colouring, and no global all-r certificate were computed. The optimized Behrend coefficient is a local improvement to this conservative budget, not a claim to improve every known multicolour lower bound. None of these results proves a conjecture by coordinate transport.

Feasible extensions. A complete implementation could maintain rational intervals for adaptive endpoints and exact squared norms, with symbolic branches at equality. An event quotient could exploit identical literal prescriptions or containment, but any pruning must prove complete coverage of all progressions. A fixed seed library could optimize $\sum_i \log N_i$ under $\prod_i r_i \le r$ using Theorem 5.1; this is conditional on genuinely cyclic seed certificates. For the sphere partition, counting actually attained shell values within each half-vector is a concrete way to seek improvement beyond the present architecture. A separate global project would be needed to turn all source O-bounds and local-lemma estimates into one explicit valid K_0.

9 Executed evidence and proof boundaries

The author-owned executable `M303-checks.py` uses Python integer and rational arithmetic only. It completed **160449 cases and 252801 assertions**. A case is a loop fixture specified below, not a theorem or an independent global parameter.

Fixture type	Cases
Separated rational lines and norm bands	1260
Sharpness and closed-endpoint witness	1
Literal-key exact probability patterns	13452
Integer colour-budget optimizations	4145
Complete small interval three-progressions	113220
Original small cyclic-seed progressions	18
Mixed-base cyclic progressions	11340
Periodically extended integer progressions	17010
Conditional-expectation choices	3

The shell fixtures use rational steps at least one; the probability fixtures include conflicting repeated keys and two rational flip probabilities. Colour-budget fixtures include every $4 \le r \le 4096$ and powers $2^{13}, \dots, 2^{64}$, testing maximality at each admissible dimension. Interval tests cover $2 \le b \le 8$, $1 \le s \le 3$, with every three-progression in each interval. The mixed-base fixtures use bases $3, 4$ in four specified products and lengths $k = 3, 4, 5$. They include every initial residue and nonzero cyclic difference, and every integer progression in the periodic extension. These are small exact checks of the written proofs, not a replay of the source's large-k construction. Parent checks are independent and excluded from these counts.

The read ledger contains pre- and post-read SHA256 hashes, actual literal intervals and unread complements. Public scope is all thirteen cached texts; bibliography reading is not reading every cited proof. Private scope is TR-item-233-astra lines 1–839 and original TR250 lines 1–64, 121–272, 721–841 and 1124–1174 (388 original lines). The original clearance chapter was located and hashed but not freshly read or used for this report; it receives no reading credit. The rest of the private corpus is unread here. The parent source-identity record authenticates cached bytes against the captured October 8 snapshot, not a live branch or a mathematical proof. No confidential material was sent to a public search.

The mathematical change to the source is a sharpening of a valid bound, not a repair of a discovered false theorem. Retained safeguards include prime-power versus prime distinctions, actual label locations, full sigma-algebra independence, half-open cuts, sign arrangements with equality, fixed outer colours before flips, and a common k-threshold before unbounded colour counts. The assessment is L3 applicability and C3 local contribution on separate scales. No score is assigned for merely renaming classical identities or executing finite fixtures.

References

[1] OpenAI mathematical preprint collection, *Quantitative Superexponential Bounds for van der Waerden Numbers*, September 23, 2026. Assigned cached manuscript, scientific sections 01–08; pinned repository snapshot recorded in the read/provenance ledger. Source claims distinguished from local conclusions throughout.

[2] J. Fox and Z. Hunter, *Three-color van der Waerden numbers grow super-exponentially*, 2026. https://arxiv.org/abs/2606.02541. Primary Theorems 1–2 and least-nonzero-digit discussion inspected; no full-paper independent proof audit.

[3] M. Campos, J. Fox and C. Schildkraut, *A new lower bound for two-color van der Waerden numbers*, 2026. https://arxiv.org/abs/2608.20824. Primary Theorem 1 and Lemma 2/proof inspected; external number-theory inputs not independently checked.

[4] H. Cao, *An improved lower bound for the van der Waerden number* $w(3, k)$, September 25, 2026. https://arxiv.org/abs/2609.31798. Abstract and metadata only; off-diagonal regime.

[5] F. A. Behrend, On sets of integers which contain no three terms in arithmetical progression, *Proc. Natl. Acad. Sci. USA* **32** (1946), 331–332. https://doi.org/10.1073/pnas.32.12.331. Classical sphere attribution; the needed partition proof is given here.

[6] M. Subbarao, *Van der Waerden Numbers and Sparse Forcing Sets*, TR-item-233-astra, confidential local comparison, current file read in full. Property-B and finite-certificate comparisons, not an original proof of the assigned superexponential construction.

[7] M. Subbarao, *Compatible Variational Ram Panels*, Technical Report TR250, September 2026, original local corpus file. Fresh passages: spaces and coordinate transfer; constrained acceptance; exact finite feasibility and metric restrictions. Read intervals recorded above; remaining portions not audited here.